

## CHISINEU-CRIS, Romania

WMO: 151360

Lat: 46.5186N

Lon: 21.5417E

Elev: 319

StdP: 14.53

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 12.4       | 16.8 | 8.9                           | 8.8 | 14.8 | 13.6 | 11.1 | 19.2 | 23.6                  | 27.0 | 21.0 | 30.7 | 5.1                   | 0    | 0.474 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 24.0                   | 93.2            | 69.0 | 90.2 | 69.4 | 87.1 | 68.6 | 73.5                | 83.5 | 72.0 | 83.4 | 70.7 | 82.1 | 6.5                  | 160  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 71.1                            | 116.1 | 76.2 | 69.2 | 108.4 | 74.9 | 67.4 | 102.0 | 73.2 | 37.3          | 82.8 | 35.9 | 83.6 | 34.7 | 82.7 | 81.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| 19.3              | 17.0 | 14.5 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |      | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |      | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 53.0   | 30.5  | 37.0  | 43.0  | 53.0  | 62.5  | 69.2  | 71.1  | 72.3  | 63.3  | 53.1  | 44.7  | 35.4 |
|                                                                               | DBStd       | 15.95  | 8.31  | 8.41  | 7.93  | 8.44  | 6.18  | 5.80  | 5.38  | 6.05  | 7.40  | 7.81  | 8.01  | 7.71 |
|                                                                               | HDD50       | 1997   | 605   | 366   | 238   | 68    | 3     | 0     | 0     | 0     | 1     | 61    | 200   | 455  |
|                                                                               | HDD65       | 5083   | 1069  | 783   | 681   | 364   | 120   | 24    | 10    | 9     | 122   | 373   | 610   | 918  |
|                                                                               | CDD50       | 3090   | 0     | 3     | 22    | 157   | 390   | 576   | 655   | 691   | 398   | 156   | 40    | 2    |
|                                                                               | CDD65       | 705    | 0     | 0     | 0     | 3     | 42    | 151   | 201   | 235   | 69    | 4     | 0     | 0    |
|                                                                               | CDH74       | 7977   | 0     | 0     | 1     | 95    | 545   | 1601  | 2190  | 2692  | 773   | 80    | 0     | 0    |
|                                                                               | CDH80       | 3125   | 0     | 0     | 0     | 6     | 112   | 576   | 883   | 1274  | 270   | 4     | 0     | 0    |
| Wind                                                                          | WSAvg       | 5.8    | 6.1   | 7.1   | 7.1   | 6.9   | 6.2   | 5.1   | 5.0   | 4.9   | 4.8   | 4.9   | 5.8   | 6.4  |
| Precipitation                                                                 | PrecAvg     | 23.3   | 1.4   | 1.2   | 1.3   | 1.9   | 2.2   | 2.8   | 2.7   | 1.9   | 2.4   | 2.1   | 1.9   | 1.8  |
|                                                                               | PrecMax     | 33.3   | 3.1   | 3.6   | 5.7   | 3.5   | 5.4   | 7.4   | 5.6   | 4.1   | 5.6   | 4.9   | 4.4   | 4.1  |
|                                                                               | PrecMin     | 12.2   | 0.3   | 0.0   | 0.1   | 0.7   | 0.4   | 0.2   | 0.7   | 0.0   | 0.7   | 0.0   | 0.0   | 0.0  |
|                                                                               | PrecStd     | 5.3    | 0.7   | 0.8   | 1.2   | 0.8   | 1.2   | 1.7   | 1.5   | 1.2   | 1.1   | 1.3   | 0.9   | 1.0  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 53.7  | 59.9  | 70.9  | 80.7  | 86.3  | 92.7  | 96.4  | 98.1  | 92.1  | 79.9  | 69.3  | 57.7 |
|                                                                               |             | MCWB   | 48.9  | 50.4  | 55.7  | 62.1  | 65.0  | 70.4  | 70.8  | 69.8  | 67.5  | 62.5  | 57.1  | 51.1 |
|                                                                               | 2%          | DB     | 48.5  | 56.1  | 65.0  | 77.0  | 83.2  | 89.8  | 92.1  | 94.1  | 86.6  | 76.1  | 64.8  | 51.9 |
|                                                                               |             | MCWB   | 45.2  | 50.0  | 52.7  | 60.3  | 65.0  | 71.1  | 70.2  | 68.9  | 67.1  | 60.9  | 55.3  | 48.0 |
|                                                                               | 5%          | DB     | 44.8  | 52.3  | 59.9  | 73.0  | 80.2  | 86.7  | 88.4  | 91.2  | 82.6  | 72.4  | 61.1  | 48.2 |
|                                                                               |             | MCWB   | 43.3  | 47.6  | 49.5  | 57.9  | 63.9  | 69.6  | 68.9  | 68.5  | 64.7  | 59.3  | 53.6  | 45.3 |
|                                                                               | 10%         | DB     | 41.3  | 49.1  | 55.6  | 68.9  | 76.6  | 83.3  | 84.8  | 87.5  | 78.4  | 67.2  | 57.0  | 45.4 |
|                                                                               |             | MCWB   | 40.1  | 45.6  | 47.4  | 55.6  | 62.7  | 68.2  | 67.6  | 68.7  | 64.3  | 57.4  | 51.7  | 43.6 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 50.0  | 53.2  | 57.2  | 65.0  | 70.3  | 76.8  | 74.7  | 75.1  | 70.9  | 65.2  | 58.2  | 51.3 |
|                                                                               |             | MCDB   | 53.5  | 58.1  | 68.1  | 75.5  | 76.1  | 86.6  | 82.2  | 86.0  | 86.3  | 74.8  | 66.2  | 56.5 |
|                                                                               | 2%          | WB     | 46.1  | 50.6  | 53.9  | 62.3  | 68.1  | 73.6  | 72.9  | 73.0  | 68.5  | 62.1  | 56.4  | 48.4 |
|                                                                               |             | MCDB   | 48.0  | 54.4  | 61.9  | 73.3  | 79.0  | 82.1  | 83.0  | 85.6  | 81.7  | 72.1  | 63.0  | 51.5 |
|                                                                               | 5%          | WB     | 43.4  | 48.2  | 51.1  | 59.8  | 66.1  | 71.6  | 71.6  | 71.4  | 66.9  | 60.5  | 54.5  | 45.8 |
|                                                                               |             | MCDB   | 44.9  | 52.7  | 58.6  | 70.3  | 76.2  | 81.7  | 83.5  | 84.5  | 78.4  | 69.9  | 60.6  | 48.0 |
|                                                                               | 10%         | WB     | 40.2  | 45.9  | 48.7  | 57.5  | 64.4  | 69.9  | 70.2  | 70.0  | 65.2  | 58.6  | 52.2  | 43.7 |
|                                                                               |             | MCDB   | 41.5  | 48.9  | 54.1  | 65.8  | 73.5  | 79.8  | 81.8  | 82.4  | 74.8  | 66.0  | 56.6  | 45.4 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 10.2  | 13.7  | 19.0  | 22.3  | 21.5  | 21.8  | 23.3  | 24.0  | 21.2  | 20.9  | 15.3  | 9.2  |
|                                                                               |             | MCDBR  | 13.9  | 19.1  | 25.5  | 28.4  | 27.4  | 26.4  | 28.7  | 29.5  | 28.2  | 29.3  | 20.7  | 13.8 |
|                                                                               |             | MCWBR  | 10.3  | 13.3  | 14.7  | 13.3  | 12.1  | 10.1  | 9.6   | 8.0   | 9.5   | 14.9  | 11.9  | 10.2 |
|                                                                               | 5% WB       | MCDBR  | 13.3  | 17.3  | 22.9  | 25.1  | 23.9  | 23.1  | 24.7  | 24.9  | 24.3  | 25.1  | 18.9  | 13.4 |
|                                                                               |             | MCWBR  | 10.3  | 12.2  | 13.6  | 12.5  | 12.2  | 10.4  | 9.6   | 8.4   | 9.4   | 13.3  | 11.3  | 10.0 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.338  | 0.365 | 0.403 | 0.459 | 0.441 | 0.439 | 0.454 | 0.457 | 0.428 | 0.390 | 0.362 | 0.338 |      |
|                                                                               | taud        | 2.323  | 2.267 | 2.197 | 2.056 | 2.175 | 2.216 | 2.178 | 2.170 | 2.213 | 2.308 | 2.337 | 2.344 |      |
|                                                                               | Ebn at Noon | 233    | 251   | 257   | 252   | 263   | 264   | 257   | 251   | 246   | 238   | 222   | 219   |      |
|                                                                               | Edh at Noon | 25     | 32    | 39    | 49    | 45    | 44    | 45    | 44    | 38    | 30    | 24    | 22    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 351    | 620   | 1018  | 1468  | 1780  | 1942  | 1913  | 1728  | 1208  | 804   | 459   | 271   |      |
|                                                                               | RadStd      | 40     | 95    | 140   | 139   | 167   | 143   | 151   | 162   | 132   | 87    | 42    | 39    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |      | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |      | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j)  | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A  | N/A         | N/A   | N/A   |       |
| (47) | Regional (60 neighbors) | +1.26 | N/A     | N/A    | +1.82 | N/A     | N/A   | -225 | -338        | +203  | +104  |       |

Nomenclature: See separate page